

Vectors

CHAPTER 1

Vectors — what they are, adding, scaling

The building block: everything in linear algebra is made of these.

THE IDEA

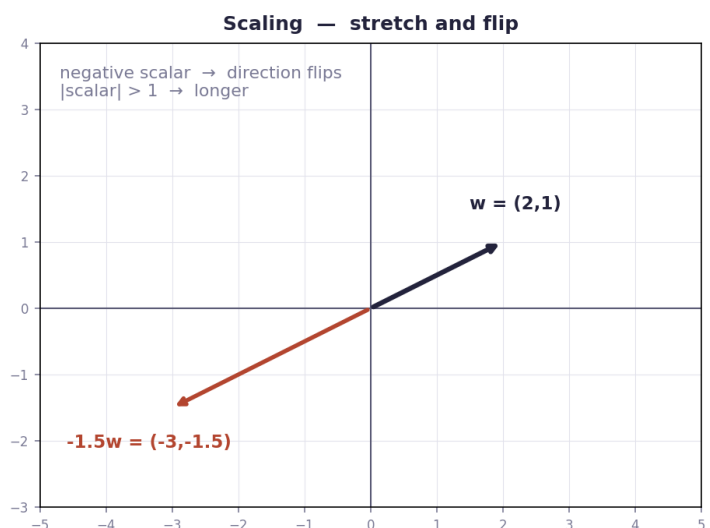
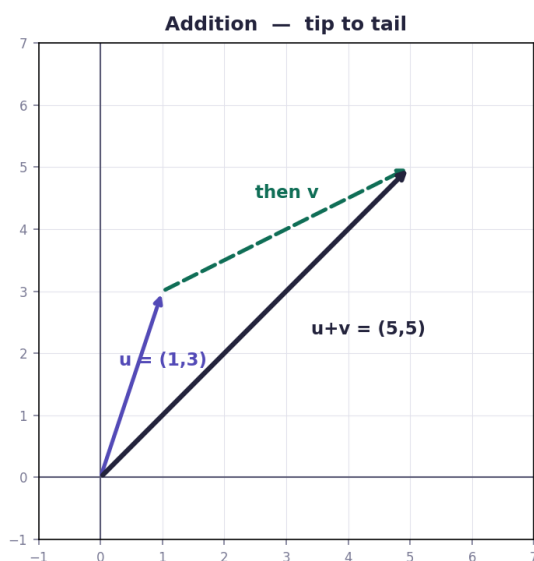
A vector is **not a point** — it is a **movement instruction**. The vector $(3, 6)$ means "go 3 right and 6 up". Because it is only an instruction, you may draw it starting anywhere in the plane and it is still the same vector. Drawing it from the origin is just a convention called **standard position**, useful because then the tip's coordinates equal the vector's numbers.

THREE WAYS TO SEE ONE VECTOR

View	What it means
Arrow	A direction and a length drawn on the grid
List of numbers	$(3, 6)$ — easy to compute with
Both	The two views are the same object. Geometry gives meaning, numbers give calculation

THE TWO OPERATIONS

Operation	How	What it does
Addition	Add matching components	Walk the first vector, then the second — tip to tail
Scaling	Multiply every component	Stretch or shrink. Negative scalar flips the direction



TAIL, TIP AND THE THREE FORMULAS

Known	Formula
tail + vector	tip = tail + vector

tail + tip

$$\text{vector} = \text{tip} - \text{tail}$$

tip + vector

$$\text{tail} = \text{tip} - \text{vector}$$

tail = the start (flat end) · **tip** = the end (arrowhead). Always verify a drawing with $\text{tip} - \text{tail}$: it must give back the vector.

UNIT VECTORS AND MAGNITUDE

Magnitude $\|v\|$ is the length: for $v = (3, 4)$, $\|v\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$. A **unit vector** has length exactly 1 — divide each component by the magnitude: $(3/5, 4/5)$. Check it: $(3/5)^2 + (4/5)^2 = 25/25 = 1$ ✓

WORKED EXAMPLES

CHAPTER 1 — WORKED EXAMPLES

ADDITION

$$u = (1, 3) \quad v = (4, 2)$$

$$u + v = (1+4, 3+2) = (5, 5)$$

SCALING + SUBTRACTION (write as adding a negative)

$$a = (2, -1) \quad b = (-3, 4)$$

$$3a = (6, -3)$$

$$-2b = (6, -8)$$

$$\begin{aligned} 3a - 2b &= (6, -3) + (6, -8) \\ &= (12, -11) \end{aligned}$$

NEGATIVE SCALING

$$w = (2, 1) \quad -1.5w = (-3, -1.5)$$

direction FLIPS, length $\times 1.5$

RULE: never subtract — turn it into adding a negative first.

The habit that prevents most errors: never subtract directly. Turn every subtraction into *adding a negative* and write that line down before combining.

TEST — Chapter 1

1. $u = (1, 3)$, $v = (4, 2)$. Compute $u + v$. → **(5, 5)**

2. $a = (2, -1)$, $b = (-3, 4)$. Compute $3a - 2b$. $\rightarrow (12, -11)$

3. $w = (2, 1)$. Compute $-1.5w$, and state the effect. $\rightarrow (-3, -1.5)$; direction flips, length $\times 1.5$

FORMULAS FROM THIS CHAPTER

Addition	$(a, b) + (c, d) = (a+c, b+d)$
Scaling	$k(a, b) = (ka, kb)$
Tip	tip = tail + vector
Vector from 2 points	vector = tip - tail
Magnitude	$\ v\ = \sqrt{x^2 + y^2}$
Unit vector	$v / \ v\ $ (check: squares sum to 1)